

AI Smart Vehicle Assistance System for Fault Diagnosis and Garage Finder

1. Overview

The **AI Smart Vehicle Assistance System for Fault Diagnosis and Garage Finder** is a web-based application that helps vehicle owners during breakdowns using Artificial Intelligence (AI), Machine Learning, and GPS-based services. Users enter vehicle symptoms, and the AI predicts possible faults with a confidence level while recommending whether the vehicle is safe to drive or should be towed. The system also finds nearby garages based on location, ratings, and specialization, estimates repair costs, and provides emergency assistance for faster and safer decision-making.

2. Challenges Intended to Solve Using AI Integration

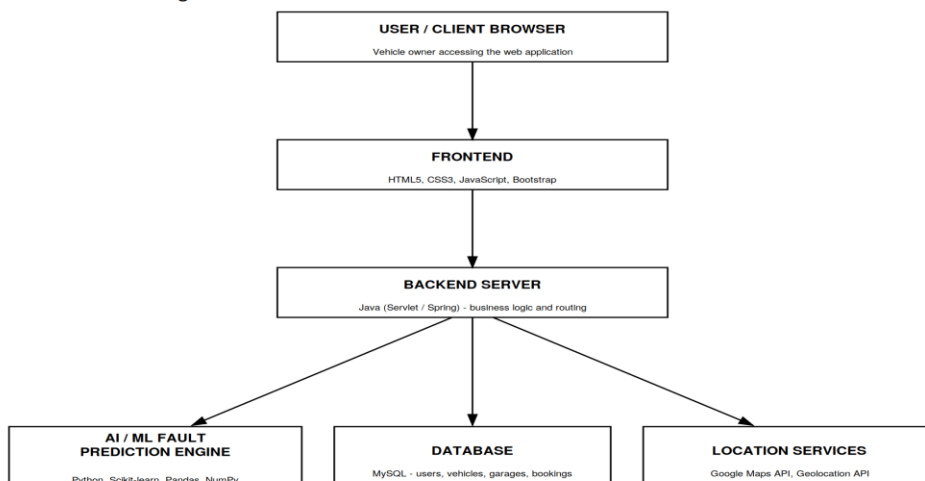
Challenges	Solutions
1. Drivers often cannot identify the cause of vehicle problems during failures.	AI analyzes user-entered symptoms and predicts possible vehicle faults with confidence levels.
2. Drivers may not know whether the vehicle is safe to continue driving after a problem occurs.	AI evaluates the severity of the fault and recommends whether the vehicle is safe to drive or requires towing.
3. Finding a reliable garage during a breakdown can be difficult, especially in unfamiliar areas.	GPS-based location services identify nearby garages and provide navigation support.
4. Users may select garages that do not specialize in their vehicle problem.	The system recommends suitable garages based on the predicted fault and repair specialization.
5. Drivers may face difficulties when immediate assistance is required during breakdown situations.	The system provides nearby garage details, emergency contacts, and location-sharing options.
6. Vehicle owners may not know the expected repair expenses before visiting a garage.	The system provides estimated repair cost ranges based on the detected vehicle problem.

3. Technology Details

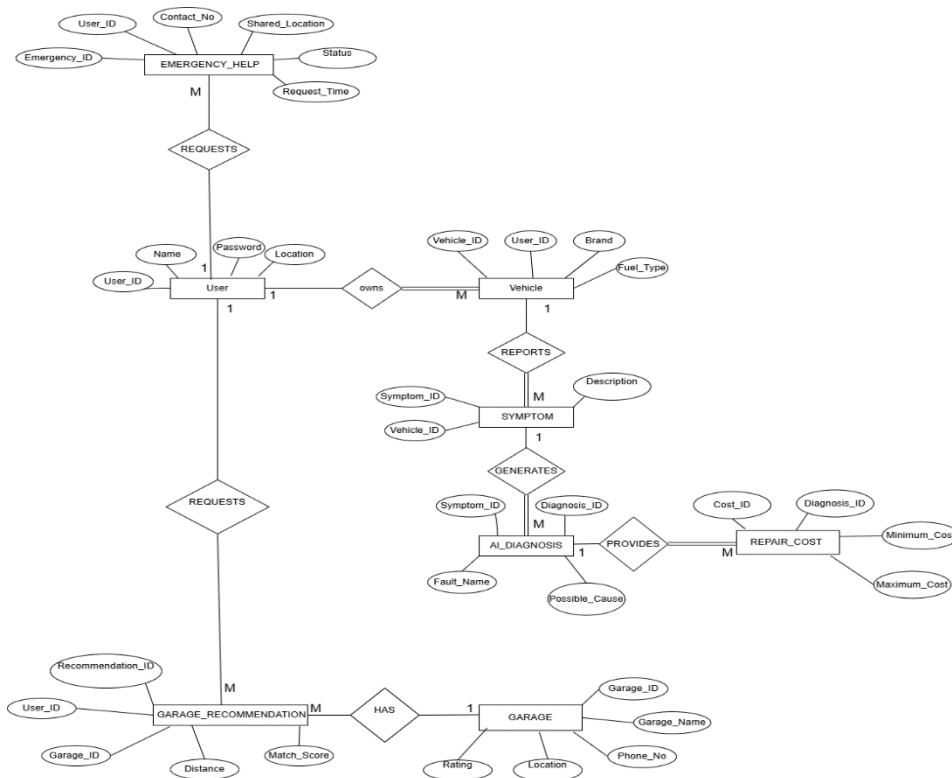
- Frontend: HTML5, CSS3, JS, Bootstrap
- Backend: Java (or Node.js)
- Database: MySQL
- AI Technologies: Python, Scikit-learn, Pandas, NumPy
- APIs: Google Maps, Geolocation
- Tools: VS Code, XAMPP, GitHub

4. System Architecture Diagram

System Architecture Diagram



5. System ER Diagram



6. Functional and Non-Functional Requirements

Functional Requirements	Non-Functional Requirements
User registration and login.	User friendly interface.
Add and manage vehicle details.	Fast response time
Enter vehicle symptoms.	Accurate AI prediction
AI-based vehicle fault prediction.	Secure user data management
Display confidence level and possible causes.	Reliable GPS tracking
Recommend whether to drive or tow the vehicle.	Responsive design
Estimate repair cost range.	High system availability
Find nearby garages using GPS.	
Rank garages based on distance, rating, and specialization.	
Provide emergency assistance and location sharing.	

7. Conclusion

The **AI Smart Vehicle Assistance System for Fault Diagnosis and Garage Finder** provides an intelligent solution for vehicle owners during breakdown situations. By combining AI-based fault detection, safety recommendations, repair cost estimation, and GPS-based garage searching, the system helps users quickly identify problems and receive suitable assistance. This project demonstrates the practical use of Artificial Intelligence to improve vehicle maintenance, reduce repair delays, and enhance road safety.

8. GitHub Link

<https://github.com/Garage-Group09/AI-Garage-Assistant>